

PROJECT REQUIREMENTS DOCUMENT

TRIBOT AI-Assisted Clinical Triage Classification & Decision Support Interface

Team: COMP9900 – W18C Donut

Client: Dr Anthony Sunjaya & MD Abu Rayhan

Organisation: TRIBOT, School of Population Health, UNSW

Term 1, 2026

1. Executive Summary

The TRIBOT Triage Module aims to design and implement an AI-assisted clinical triage classification system to support healthcare professionals in determining patient urgency levels. The solution will combine machine learning techniques with rule-based safety mechanisms to ensure clinical reliability, transparency, explainability, and usability.

2. Background & Objectives

Efficient triage is critical in healthcare settings to prioritise patient care appropriately. Inconsistent triage decisions may result in delayed treatment or resource misallocation. This project will develop a functional prototype capable of classifying symptom input into urgency levels.

- Primary Objectives:
- Develop a supervised ML-based triage classification model
- Implement a rule-based fallback safety system
- Design and implement a RESTful backend API
- Build a React-based clinician dashboard
- Provide full evaluation metrics and documentation

3. Functional Requirements

3.1 Input Handling

- The system must:
- Accept text-based symptom descriptions
- Optionally support structured fields (age, duration, comorbidities)
- Validate and sanitise all inputs
- Log submissions for auditing and evaluation

3.2 Triage Classification Engine

- The engine must include:
- A multi-class ML model (Low / Medium / High urgency)
- Confidence scoring for each prediction
- Mandatory rule-based override logic for critical symptoms

3.3 REST API Backend

- Required endpoints:
- POST /triage – Returns urgency classification + confidence score
- GET /model-metrics – Returns evaluation metrics
- GET /health – System health status

3.4 Clinician Dashboard (Frontend)

- Frontend must:
- Be implemented using React
- Provide a clean, HCI-compliant interface
- Display urgency levels using colour-coded indicators
- Show confidence score and triggered rule-based alerts

3.5 Evaluation Framework

- System must generate:
- Accuracy, Precision, Recall, F1-score
- Confusion matrix

4. Non-Functional Requirements

- Performance:
- API response time < 2 seconds
- Frontend load time < 3 seconds
- Security:
- No identifiable patient data storage
- Input validation against injection attacks
- Reliability:
- Rule-based fallback must operate independently of ML model

5. Expected Deliverables

- Functional triage prototype
- Complete source code repository
- Technical documentation & API documentation
- Model evaluation report
- Final demonstration

6. Milestone Plan (10 Weeks)

- Weeks 1–2: Requirements finalisation & system architecture
- Weeks 3–4: Dataset preprocessing & baseline model
- Weeks 5–6: Backend API development
- Week 7: Frontend integration
- Week 8: Rule-based safety implementation
- Week 9: Evaluation & optimisation
- Week 10: Final testing & demo